NVIDIA ACCELERATED COMPUTING & KUMO.AI // APPLIED STRATEGY

# Establishing Foundational Intelligence Infrastructure for Global Financial Services

Architectural Vision, 6 Core Tenets, 9 Systemic Threat Frontiers & The 5 Hard Primitives
**Scope:** NVIDIA & Kumo.ai Applied Strategy • Leadership & Engineering Roadmap

## 1. Executive Applied Strategy & Lighthouse Operating Model

Our overarching applied strategy anchors directly on a clear vision: **establishing NVIDIA and Kumo as the open-source foundational intelligence infrastructure for global financial services**. We are not positioning ourselves as a bespoke professional services group or a vendor selling brittle point solutions. Our operating model is **open-source blueprints co-developed alongside engineering lighthouses**. We build once with an advanced pioneer, open-source the core architectural primitives, and replicate that blueprint across global Tier-1 banks.

Today, traditional commercial banks maintain dozens of siloed, fragile XGBoost models fed by hundreds of brittle SQL pipelines. We are here to replace that fragmented complexity with unified foundation backbones.

The leading empirical benchmark validating this paradigm is **Revolut PRAGMA**. By training an autoregressive Transaction Foundation Model across **24 billion banking events in 111 countries** using NVIDIA NeMo, H100 Tensor Core GPUs, and RAPIDS cuDF, Revolut transformed their bottom line: they achieved a **+130% PR-AUC uplift** in credit risk underwriting, a **+65% increase** in fraud detection recall, and a **+41% gain** in recommendation precision. Crucially, Revolut proved that a single pre-trained behavioral representation backbone beats dozens of hand-crafted feature stores, eliminating the 80% feature engineering tax that paralyzes bank data science teams.

**The Critical AML Vulnerability & The 47% Collapse:**
When Revolut evaluated one-dimensional sequence foundation models on Anti-Money Laundering (AML) benchmarks, performance **collapsed by 47%**.
*Why did it collapse?* Because money laundering, mule rings, and authorized push payment scams are not sequence problems—they are **topological problems**. A 1D transformer models an individual account's personal transaction history over time. But when an organized syndicate moves stolen funds, they deliberately structure transactions below reporting thresholds. Every individual transaction looks statistically benign. The criminal signal exists entirely in the counterparty graph: the rapid fan-out to five mule accounts, the cyclic looping across shell companies, and the rapid fund dissipation.
A sequence model is blind to network topology, while a traditional graph neural network is blind to temporal customer journeys. Our core modeling breakthrough must be the **Relational Transaction Foundation Model (RTFM)**—fusing sequence encoders with relational graph attention into a single inductive backbone.

That is how we replicate Revolut's win across the global banking tier:
• **Revolut:** Expand from 1D transaction sequences to multi-hop Graph Foundation Models to dismantle complex mule networks.
• **Intesa Sanpaolo:** Bring onto this blueprint to stop real-time authorized push payment and synthetic voice scams.
• **Banco Santander:** Transition off bespoke, hand-crafted GNNs onto unified Kumo Relational Foundation Models.
• **NPCI (India):** Deliver the sub-50ms streaming engine needed to score 15 billion monthly UPI transactions.

As Chief Scientist Jure Leskovec framed: *'The less is there, the better.'* We do not need a sprawling software catalog. We need a minimal, composable set of foundation primitives that any institution can deploy inside their private Snowflake or Databricks lakehouse with zero data egress.

## 2. The 6 Foundational Tenets Guiding Platform Development

**Tenet 1: Fraud is an Identity Problem Upstream, Not a Payment Problem Downstream**

When an authorized push payment scam, pig-butchering attack, or synthetic ATO occurs, the payment authorization is just the crime scene. The legitimate account holder authenticates the transaction willingly. Looking strictly at the payment transaction produces zero anomalies. The attack originated days or weeks earlier in credential compromises, social engineering, and sleeper accounts. If our models only evaluate the payment edge, we are scoring the symptom rather than intercepting the syndicate.

**Tenet 2: Focus Relentlessly on Hard Foundational Primitives**

Build what only NVIDIA and Kumo can solve, and ruthlessly reject self-fulfilling prophecies. This ties directly to Jensen Huang's core mission for NVIDIA: 'pioneering accelerated computing to tackle impactful challenges no one else can solve.' We are not here to build commodity feature stores, lightweight dashboards, or thin wrappers around generic LLMs. Our mission is solving mathematically and computationally intractable challenges by combining accelerated computing with full-stack AI: scaling dynamic graph memory across 100 billion edges, zero-shot relational transfer learning, and sub-30ms inductive serving on GPUs and Tensor Cores.

**Tenet 3: Zero Critical Gaps in Our Primitive Layer**

If our platform cannot handle sub-30ms card switches or chokes on massive multi-hop graph traversals, banks will be forced to buy point solutions. Our primitives must natively cover the entire surface area without functional voids.

**Tenet 4: Meet Banks Where They Are — With Autonomous Agents**

Tier-1 banks are deeply entrenched in Apache Flink, Apache Kafka, Snowflake, Databricks, and complex relational schemas. We cannot operate three years orthogonally and ask them to rewrite their data infrastructure. Our autonomous coding agents are the pragmatic bridge: they ingest customer schemas directly, map them into PQL, and deploy containerized, zero-data-egress NIMs inside their VPCs to deliver quantifiable P&L; impact within a 14-day shadow POC.

**Tenet 5: 'The Less Is There, The Better'**

We reject bloated, monolithic enterprise suites in favor of lean, composable building blocks that drop directly into existing bank data pipelines.

**Tenet 6: Closed-Loop Symbiosis (Human-in-the-Loop & AI-in-the-Loop)**

Banks face an operational paradox: manual human review creates an unbearable bottleneck on instant rails, while unconstrained automated ML triggers severe regulatory pushback under Federal Reserve SR 11-7. We enforce a closed-loop symbiosis: wherever there is Human-In-The-Loop, deploy AI-In-The-Loop agents to automate close to 100% of repetitive investigation volume; wherever there is automated ML, maintain continuous Human-In-The-Loop validation to achieve 100% quality coverage.

**Upstream Fleet of Agents:** In our architecture, autonomous agents bat UPSTREAM ahead of foundation models:
1. **Coding Agents:** Automate schema onboarding and compile natural language into PQL queries.
2. **Customer Experience Agents:** Fuse multi-modal inputs (video IDV liveness, document OCR, card images) with graph topology to eliminate friction for 98% of legitimate customers.
3. **Risk Ops Agents:** Investigate multi-hop subgraphs, generate attribution paths, and file automated SARs.

## 3. The 9 Existential Threat Frontiers (2026–2028 Total Space of Fraud)

Mapping every existential threat frontier into a rigorous 6-dimension taxonomy: SLA, Anomaly Type, Model Engine, Technology Backbone, Lighthouse Target, and Business Impact. There is zero overlap and zero blind spots.

| Frontier / Threat | SLA & Anomaly Type | Model & Backbone | Lighthouse Target & Business Impact |
|---|---|---|---|
| **1. Card & Payment Auth** (CNP / Point of Sale) | <30ms Inline Sequence Deviation | Transaction FM (PRAGMA) NeMo, cuDF, H100 | **JPMC / Santander** +65% Fraud Recall; -$42M gross losses |
| **2. Instant Rails & Mules** (FedNow, RTP, UPI) | <50ms Inline Streaming Graph Flow | Sub-15ms Subgraph Framer cuGraph-Stream, Triton | **NPCI (UPI) / Chime** Dismantles mule networks across 240k peak TPS |
| **3. Account Takeover (ATO)** (Credential stuffing, SIM swap) | <100ms Session Behavioral Baseline Shift | KumoTabular + Diffusion Kumo-TS (Tesseract) | **Citizens Bank** -88% ATO claims; saves $18M ARR |
| **4. APP Scams & Romance** (Authorized Push Payment) | <200ms Step-Up Relational Counterparty Deception | Kumo Relational FM (RFM) Snowflake SPCS, PQL | **Intesa Sanpaolo** Catches overseas mule clusters willingly paid |
| **5. Synthetic Identity Fraud** (Deepfakes, fabricated SSNs) | Batch & Onboarding Multi-Modal Identity Collapse | Multi-Modal Tokenizer PyTorch Frame + Vision | **Fifth Third / TD Bank** -92% credit bust-out losses across sleeper rings |
| **6. Cross-Border AML** (Layering, shell companies) | T+1 Daily Ingestion Cyclic Multi-Hop Topology | Relational Trans FM (RTFM) Path-Centric Flow Kernel | **Global Tier-1 G-SIBs** Overcomes PRAGMA's 47% AML performance deficit |

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|---|---|---|---|
| **7. Merchant Bust-Out Rings** (Credit stacking, fake volume) | Near-Real-Time Bipartite Graph Collusion | Kumo 3 Dynamic GNN WholeGraph + NVLink | **Merchant Acquirers** Identifies coordinated merchant default syndicates |
| **8. Model Governance & Bias** (SR 11-7, EU AI Act Art. 14) | Audit & Compliance Future-Data Target Leakage | Deterministic ANCHOR_TIME Cryptographic Audit Fabric | **Federal Reserve / ECB** 100% auditable attribution; zero data leakage |
| **9. Multi-Agent Attack Swarms** (Autonomous fraud botnets) | Dynamic Real-Time Coordinated Adversarial Blitz | Autonomous Defense Fleet NeMo Agents + MCP | **Sovereign Financial Rails** Automated adversarial red-teaming & defense |

## 4. The 5-Layer RTFM Stack & The 5 Hard Frontier Primitives

Our production foundation is strong: Kumo 3 containerized NIMs on NGC, Kumo RFM zero-shot relational transformers, KumoTabular with PyTorch Frame, the Financial Fraud Blueprint on Dynamo-Triton, cuGraph-PyG, WholeGraph, DiskGraph, and Confidential Computing enclaves on Hopper and Blackwell. To bridge this asset base to real-time banking operations, we deliver the unified 5-layer stack and 5 hard primitives:

| Layer | Architecture Component | Technical Function & Execution |
|---|---|---|
| **Layer 5** | **Governance & Autonomous Agents** | Autonomous Agent Fleets + SR 11-7 Cryptographic Audit Fabric & Topological Subgraph XAI. |
| **Layer 4** | **Real-Time Serving** | TensorRT-GNN (<10ms) + Temporal Positive-Unlabeled (PU) Label-Delay Calibration Engine. |
| **Layer 3** | **Frontier Models** | Relational Transaction Foundation Model (RTFM) + Path-Centric Flow Kernel + Generative Simulator. |
| **Layer 2** | **Data Fabric** | Point-in-Time Arrow IPC Sampler + Streaming cuGraph-ER + Multimodal ISO 20022 Tokenizer. |
| **Layer 1** | **Accelerated Compute** | NVLink Subgraph Memory Cache + Confidential GPU TEE Enclaves (Hopper/Blackwell). |

### The 5 Hard Frontier Primitives We Must Build:

- **Primitive 1: Sub-15ms Dynamic Subgraph Framing Kernel (cuGraph-Stream):** cuGraph-PyG executes offline batch sampling, but instant payment rails (FedNow, UPI) clear in under 30 milliseconds. cuGraph-Stream ingests streaming payment events directly from Kafka into GPU memory via GPUDirect RDMA, extracting the local 2-hop temporal neighborhood in under 8 milliseconds.
- **Primitive 2: WholeGraph Distributed Tiered Memory (100B+ Dynamic Edges):** While WholeGraph was open-sourced and Kumo built DiskGraph for NVMe scratch, WholeGraph is not yet wired into KumoRFM. At NPCI scale (15B monthly UPI transactions across 100B edges), single-GPU memory fails. We pool memory over NVLink: caching hot nodes in HBM, mapping neighborhoods in host RAM, and paging historical edges from NVMe

without CPU stalls.

- **Primitive 3: Multimodal Schema-Agnostic Tokenizer (ISO 20022 Enriched Payloads):** Financial transactions are no longer flat numerical rows. Under ISO 20022, they carry rich XML payloads, invoice memos, call audio transcripts, and video IDV liveness. We need a unified foundation tokenizer encoding structured tabular cells, Fourier-projected continuous amounts, XML tags, audio transcripts, and video into a shared latent space.
- **Primitive 4: Inductive Subgraph Inference Kernel (TensorRT-GNN on Triton):** KumoRFM is mathematically inductive, but today serving is warehouse-bound at second-level latencies. We compile KumoRFM into a standalone TensorRT-GNN inference engine deployed inside Triton to serve inductive predictions on Tensor Cores in under 100 milliseconds.
- **Primitive 5: Temporal Edge Pruning Operator and Label-Delay Engine:** Kumo solved temporal cutoffs in software with declarative ANCHOR_TIMESTAMP, but software filtering causes GPU sync stalls during live streaming. We need a low-level CUDA operator masking future edges at warp speed for 100% SR 11-7 compliance. Furthermore, real banking operations suffer 30-to-90 day chargeback confirmation lags. Our Positive-Unlabeled (PU) learning engine models this delay, preventing unflagged recent transactions from corrupting model weights during online adaptation.